College of Engineering

COMP 430 Data Privacy and Security

Final Project Report

Project Title: Can LLMs Pass COMP 430?

Group Members: Selin Başkurt, Barış Toprak, Betül Rana Çetin, Mehmet Batu Hayırlıoğlu

Table of Contents

I. Abstract
II. Introduction
III. Overall Project Structure
    Components
IV. Experimental Design
    A. Exam Construction
    B. LLMs Under Evaluation
    C. Prompting Conditions
    D. Retrieval-Augmented Generation (RAG)
V. Methods
    A. Execution Pipeline (Jupyter Notebook)
    B. Question Distributions
    C. Scoring and Grading
    D. Controlled Variables
VI. Analysis and Results
    A. Overall Performance
    B. Performance by Question Type
    C. Performance by Topic
        MT1 Performance by Topic (Non-RAG vs RAG)
        MT2 Performance by Topic (Non-RAG vs RAG)
    D. Performance by Difficulty
    E. Effect of RAG
VII. Discussion
VIII. Limitations
IX. Conclusion
References

I. Abstract

Large Language Models (LLMs) are increasingly used as assistant and problem solver tools for homeworks and projects in higher education. Despite their popularity, their reliability for subjects such as data privacy and security remains uncertain. This project systematically evaluates and shows whether modern LLMs can reliably solve COMP 430 level exam questions and whether their performance can be improved using prompt engineering and retrieval-augmented generation (RAG).

We constructed two exams covering COMP 430 course content: one including Midterm 1 topics (anonymization, differential privacy, local differential privacy) and another including Midterm 2 topics (cybersecurity, authentication, database security). Each exam includes true/false, multiple-choice, and open-ended questions labeled by topic and difficulty level. Multiple LLMs were evaluated under controlled prompting conditions using an automated Jupyter Notebook pipeline that performs grading, aggregation, and analysis.

Our results show that while modern LLMs can achieve passing-level performance on objective questions, they exhibit topic-specific weaknesses and reduced accuracy on harder and open-ended questions. Incorporating lecture slides via RAG improves accuracy in some topics. Overall, the findings highlight both the promise and the risks of using LLMs as educational tools in privacy and security contexts.

II. Introduction

Large Language Models such as GPT-family models, Gemini, and LLaMA-based systems are now widely used by students as study aids, tutoring tools, and even exam assistants. Their ability to generate complete and confident responses has led to increasing reliance on them in academic areas. However, confidence and speed do not simply imply correctness, particularly in domains where errors can have serious impacts. The COMP 430 course focuses on data privacy and security, covering topics such as anonymization, differential privacy, authentication mechanisms, and database security. These topics require precise reasoning, careful interpretation of formal definitions, and accurate application of algorithms. Mistakes in these areas can lead to incorrect system designs, security vulnerabilities, or privacy violations. Despite growing concerns about AI-assisted cheating and academic integrity, there is limited structured analysis of how well LLMs actually perform on privacy and security exam content. Existing discussions often treat AI performance without examining topic-level sensitivity, difficulty effects, or differences across question types.

The goal of this project is to fill this gap by conducting a controlled benchmarking study of multiple LLMs on COMP 430–level exam questions. We aim to answer three main

research questions: Can modern LLMs reliably pass COMP 430 exams? How does performance vary across topics, question types, and difficulty levels? Can access to course materials via retrieval-augmented generation improve performance?

III. Overall Project Structure

The project is organized as a modular evaluation pipeline that takes structured exam data as input and produces graded results and performance analytics as output.

Components

1. Exam Dataset

Two full exams were constructed:

Exam 1 (Midterm 1 Topics): Anonymization, Differential Privacy, Local Differential Privacy

Exam 2 (Midterm 2 Topics): Cybersecurity, Passwords and Authentication, Database Security

Each question is stored in JSON format with metadata fields including; exam identifier, topic, difficulty (easy, medium, hard), question type (True/False, Multiple Choice, Open-Ended).

2. LLM Evaluation Layer

Multiple LLMs are queried via APIs under identical conditions. Temperature is fixed to reduce randomness and ensure fair comparison across models.

3. Grading Engine

True/False and Multiple-Choice questions are graded automatically using answer keys.

Open-ended questions are graded using a rubric-based scoring scheme by us.

4. Analysis and Visualization Module

Aggregates scores by model, topic, difficulty, and question type, and generates tables and plots for comparison.

IV. Experimental Design

A. Exam Construction

Two exams were designed to resemble real COMP 430 midterms in format and content. Each exam contains a mix of:

- True/False questions
- Multiple-choice questions
- Open-ended questions

Each question is manually labeled with:

- Topic category
- Difficulty level (easy, medium, hard)

This labeling allows controlled analysis of how different factors influence LLM performance.

B. LLMs Under Evaluation

DeepSeek R1-0528 [3]

DeepSeek R1-0528 is a reasoning-focused large language model developed by DeepSeek. It is designed to perform well on structured problem-solving tasks, including logical reasoning, mathematical questions, and multi-step analysis. The model emphasizes accuracy and consistency in reasoning-intensive scenarios.

Meta Llama 3.1 8B Instruct [4]

Llama 3.1 8B Instruct is an instruction-tuned model released by Meta as part of the Llama 3.1 family. With approximately 8 billion parameters, it is optimized for following user instructions, generating coherent responses, and supporting general-purpose natural language tasks while remaining relatively lightweight compared to larger models.

Google Gemini 2.0 Flash Lite [5]

Gemini 2.0 Flash Lite is a lightweight variant of Google's Gemini model family. It is designed for fast inference and efficiency, prioritizing low latency and reduced computational cost. While smaller in scale, it aims to maintain reasonable performance across common language understanding and generation tasks.

NVIDIA Nemotron Nano 9B V2 [6]

Nemotron Nano 9B V2 is a compact large language model developed by NVIDIA, containing approximately 9 billion parameters. It is part of the Nemotron family and is

intended for efficient deployment, balancing performance and resource usage. The model targets use cases such as conversational AI and general language tasks.

OpenAI gpt-oss-20b [7]

Gpt-oss-20b is a 20-billion-parameter open-source language model released by OpenAI. It is designed to provide strong general-purpose language understanding and generation capabilities. Due to its larger parameter count, it is expected to perform better on complex tasks compared to smaller models, though at higher computational cost.

C. Prompting Conditions

Two prompting strategies are tested:

- **P1 (Strict Format Prompt):** Direct question with a constraint to output only the final answer without explanation.
- **P2 (Slide-Fed Prompt):** The same question augmented with retrieved lecture slide content using RAG.

D. Retrieval-Augmented Generation (RAG)

Lecture slides are converted into text, embedded, and stored in a vector database. For each question, the most relevant slide segments are retrieved and appended to the prompt. Performance with and without RAG is compared.

V. Methods

A. Execution Pipeline (Jupyter Notebook)

The experiment is implemented in a Jupyter Notebook environment using an automated evaluation pipeline. For each model and prompt condition, the notebook:

1. Loads exam JSON files, user selects which midterm to be solved.

2. Iterates over LLMs and prompting conditions (P1 and P2).

3. Sends each question to the corresponding model API and stores responses.

4. Grades True/False and Multiple-Choice answers automatically using an answer key.

5. Saves open-ended question responses for us to grade it based on our rubric.

6. Aggregates results into tables by model, topic, difficulty, and question type.

7. Generates summary plots and tables for reporting.

In RAG mode (P2), lecture slides are converted into text, embedded, and stored in a vector database. For each question, the most relevant segments are retrieved using semantic similarity search and appended to the prompt before querying the model.

B. Question Distributions

Each exam consists of 40 True/False and Multiple-Choice questions in total. For topic balance, each major topic contributes on average 5 True/False and 5 Multiple-Choice questions, ensuring uniform coverage. In addition, each exam includes open-ended questions worth a total of 20 points.

C. Scoring and Grading

For True/False and Multiple-Choice questions, the following scoring rule is applied consistently across all models and conditions:

- Correct answer: +2 points

- Incorrect answer: −0.5 points

Open-ended questions are graded using a fixed-point rubric based on: (i) correct identification of core concepts, (ii) reasoning quality, and (iii) correctness of the final conclusion. Partial credit is awarded for partially correct but incomplete answers.

An example question is written below, you can see all of our question from the json files we submitted:


```
{
 "id": 1,
 "topic": "Anonymity Notions",
 "type": "TF",
 "question": "Removing EIs (Explicit Identifiers) is necessary and alone sufficient to ensure individuals privacy.",
 "options": "True / False",
 "correct_answer": "False",
 "difficulty": "easy"
},
```

D. Controlled Variables

To ensure fairness and reproducibility:

- All models use the same temperature setting.
- Prompts follow a fixed template.
- No external browsing or tool use is enabled.
- Each exam is run independently for each model and prompt type.

VI. Analysis and Results

A. Overall Performance

Across models, objective questions (True/False and Multiple Choice) achieve higher accuracy than open-ended questions. Some models achieve passing grades, but none show high performance across all topics and difficulty levels.

A.1. Midterm 1

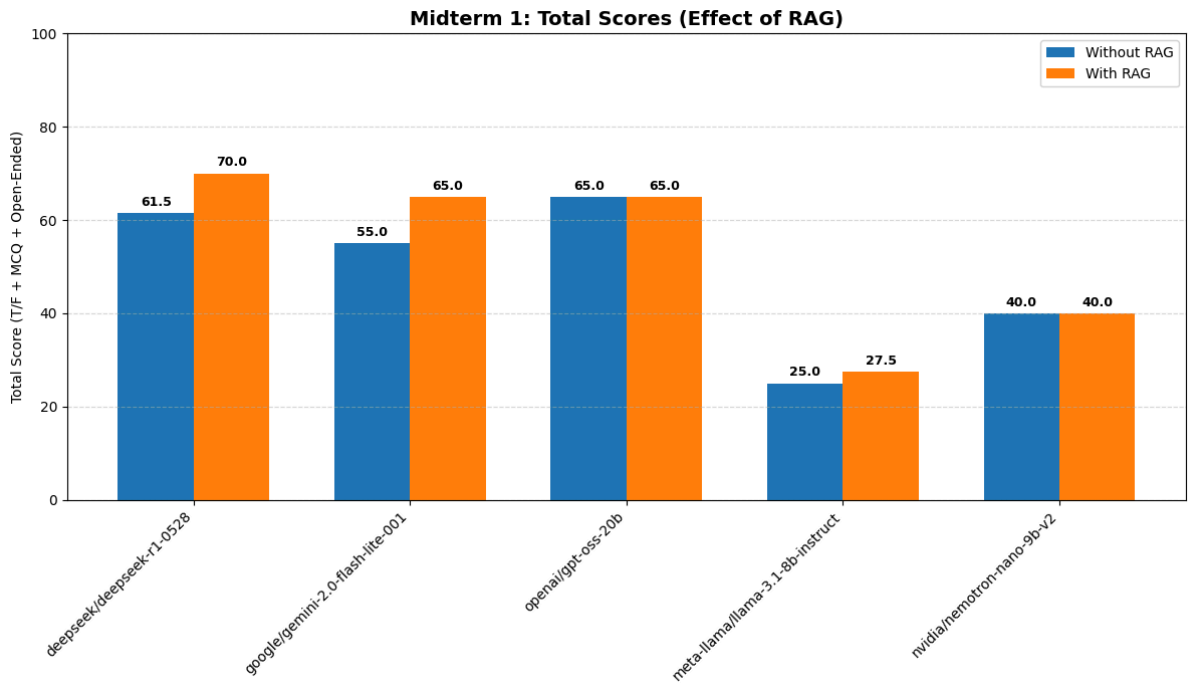

Figure 1: Midterm 1 total scores and the effect of RAG

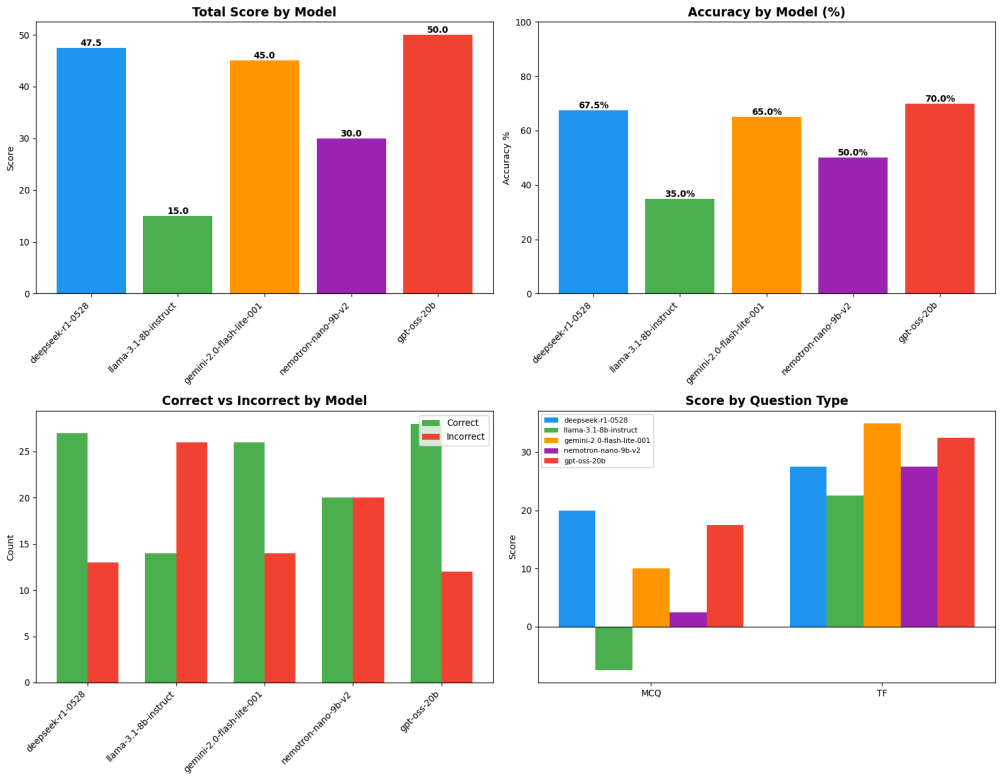

Figure 2: Overall Performance in Midterm 1

Figure 2 shows overall performance of the evaluated LLMs in Midterm 1, including total score, accuracy, correct versus incorrect responses, and performance by question type. All models were evaluated out of 100 points.

In the evaluated models, OpenAI gpt-oss-20b shows the highest performance, with a total score of 50.0, 28 correct responses, 12 incorrect responses, and an accuracy of 70.0%. This shows strong consistency across the evaluation.

DeepSeek R1-0528 was second, scoring 47.5 points with 27 correct and 13 incorrect answers. It had an accuracy of 67.5%. Its performance is after the top model, showing reasoning capability.

Google Gemini 2.0 Flash Lite scored 45.0, with 26 correct and 14 incorrect responses, equaling 65.0% accuracy. While slightly less than the top two, Gemini had stable performance.

In comparison, NVIDIA Nemotron Nano 9B V2 achieved a score of 30.0, with an equal number of 20 correct and 20 incorrect responses, resulting in 50.0% accuracy, reflecting a 50/50 performance.

The least successful results were for Meta Llama 3.1 8B Instruct, which scored only 15.0 points, with 14 correct and 26 incorrect responses, and had an accuracy of 35.0%. This model struggled significantly compared to the others.

As shown in Figure 1, all models performed better on True/False (TF) questions than on Multiple Choice Questions (MCQs). Performance differences in MCQs were a major factor contributing to the overall score gap, particularly for Llama 3.1 8B Instruct, which showed notably weak MCQ results.

Overall, the results in Figure 1 indicate that gpt-oss-20b, DeepSeek R1-0528, and Gemini 2.0 Flash Lite outperform the remaining models in terms of accuracy, consistency, and total score in Midterm 1.

A.2. Midterm 2:

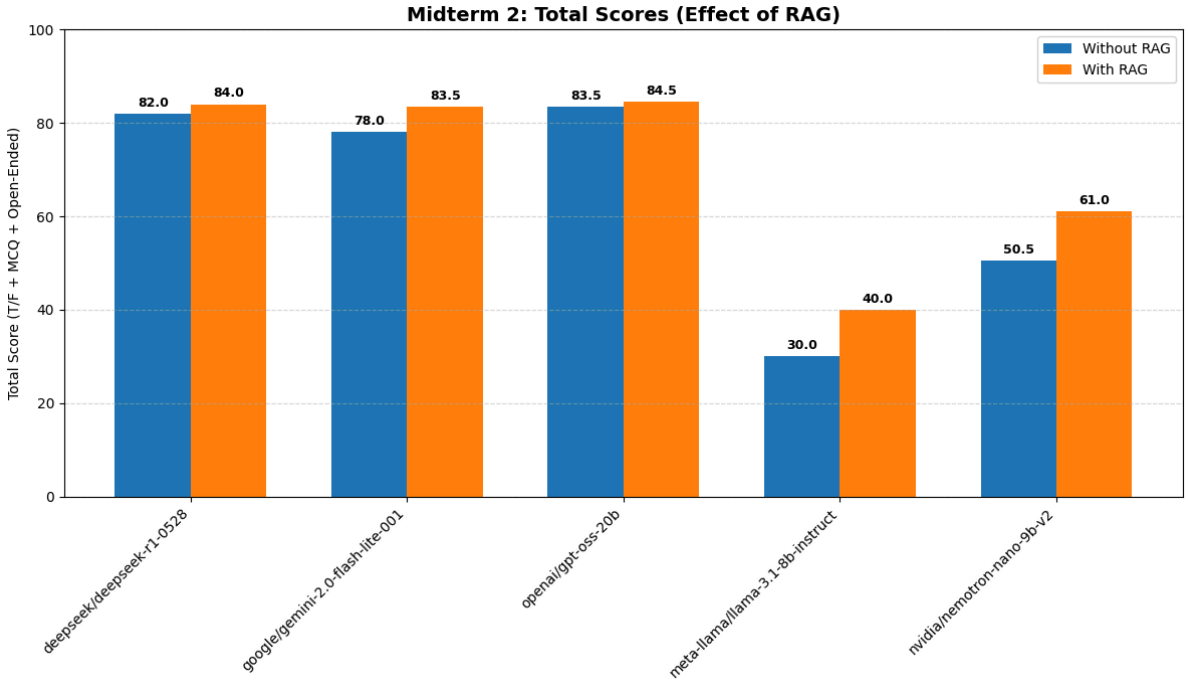

Figure 3: Midterm 2 total scores and the effect of RAG

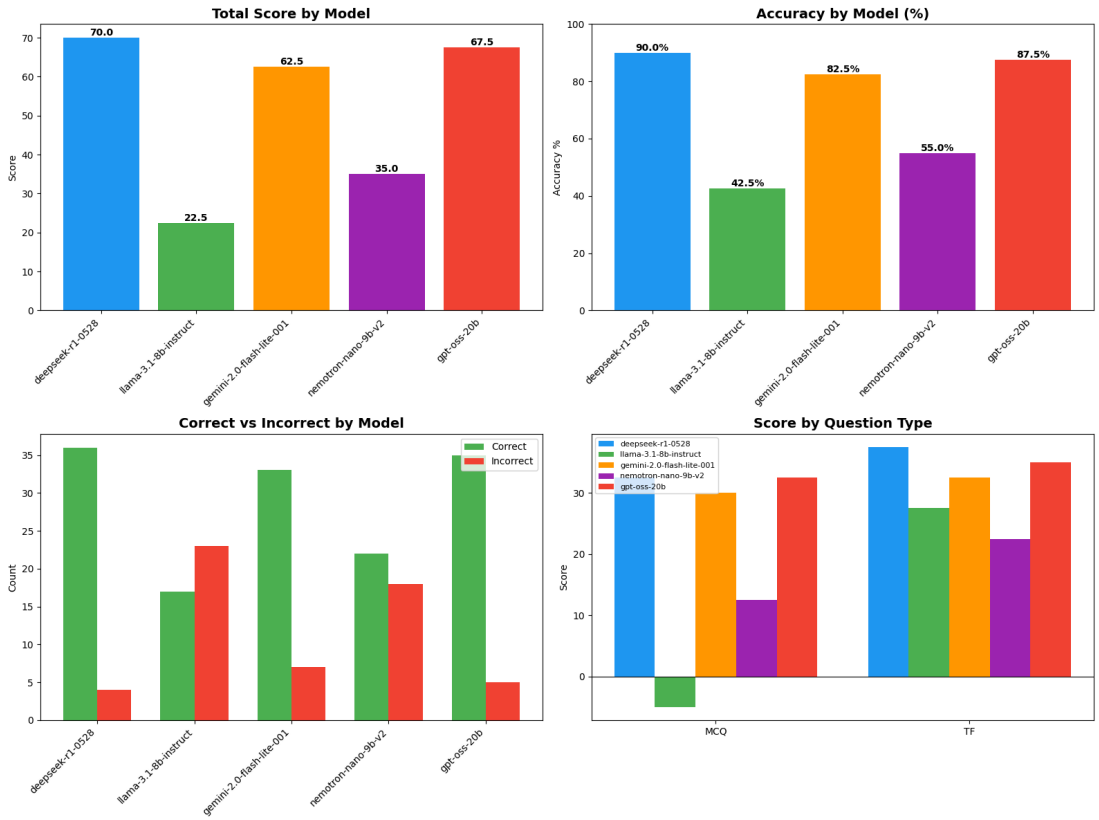

Figure 4: The Overall Performance in Midterm 2

Figure 4 shows the performance of the evaluated LLMs in Midterm 2, summarizing total scores, accuracy, correct with incorrect responses, and performance by question types (MCQ and True/False). All models were evaluated out of 100 points.

In the evaluated models, DeepSeek R1-0528 had the highest performance, winning with a total score of 70.0, with 36 correct and 4 incorrect responses, resulting in an accuracy of 90.0%. This reflects an increased improvement compared to Midterm 1 and shows strong consistency across both question types.

OpenAI gpt-oss-20b is close, scoring 67.5 points with 35 correct and 5 incorrect answers, meaning an accuracy of 87.5%. The model showed reliable performance and a low error rate throughout the evaluation.

Google Gemini 2.0 Flash Lite achieved a total score of 62.5, with 33 correct and 7 incorrect responses, yielding an accuracy of 82.5%. It is slightly behind the top two models, by maintaining strong and stable performance.

In comparison, NVIDIA Nemotron Nano 9B V2 showed a moderate score of 35.0, with 22 correct and 18 incorrect responses, resulting in 55.0% accuracy,which indicates partial improvement but continued inconsistency.

The weakest performance was again observed for Meta Llama 3.1 8B Instruct, which scored 22.5 points with 17 correct and 23 incorrect responses, achieving an accuracy of 42.5%.

As seen in Figure 4, all models performed better on True/False (TF) questions than on Multiple Choice Questions (MCQs). MCQ performance remained the main source of difference, most importantly for Llama 3.1 8B Instruct, which showed negative MCQ results, with a huge impact on its overall score.

Overall, the results in Figure 4 had a clear performance ladder in Midterm 2, with DeepSeek R1-0528, gpt-oss-20b, and Gemini 2.0 Flash Lite showing more accuracy and consistency compared to the remaining models.

A.3. Midterm 1 vs Midterm 2

While most models achieved higher scores and accuracy in Midterm 2, this improvement should be seen as stronger connection with the Midterm 2 question topics rather than overall model improvement. DeepSeek R1-0528, OpenAI gpt-oss-20b, and Google Gemini 2.0 Flash Lite demonstrated consistently okay performance across both midterms, whereas Meta Llama 3.1 8B Instruct and NVIDIA Nemotron Nano 9B V2 remained more vulnerable to changing topics.

B. Performance by Question Type

- **True/False:** Highest accuracy across all models.
- **Multiple Choice:** Moderate accuracy, with errors often arising from subtle distractors.
- **Open-Ended:** Lowest accuracy, especially on medium and hard questions requiring multi-step reasoning.

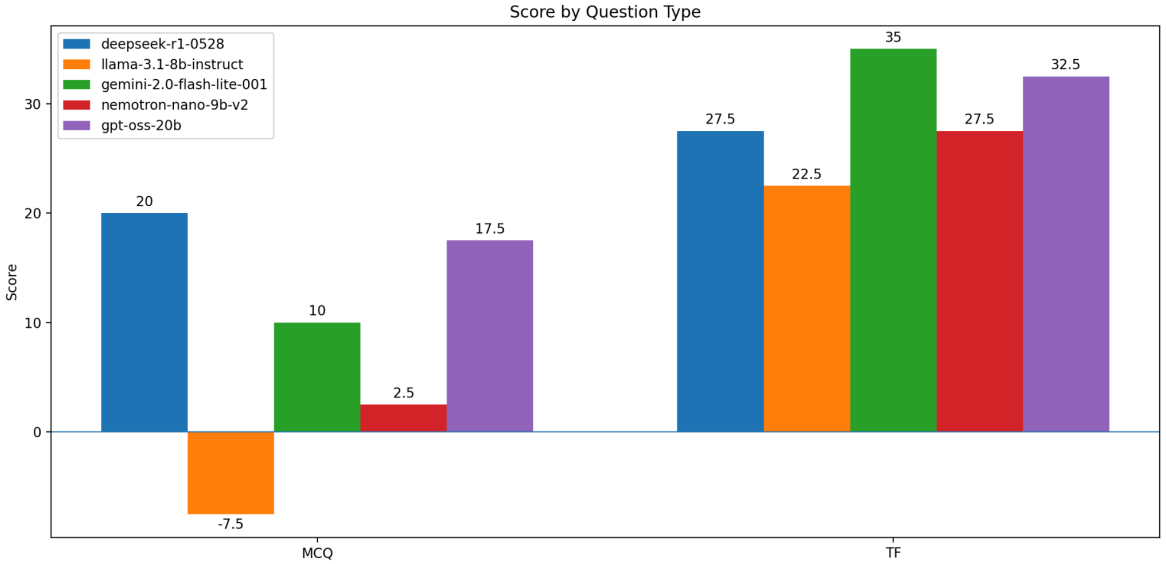

Figure 5: Score By Question Type

Figure 5 shows the performance of the evaluated LLMs by question type (MCQ and True/False). Across all models, True/False (TF) questions consistently saw higher scores than Multiple Choice Questions (MCQs), meaning that True/False decision tasks were generally easier for the models. DeepSeek R1-0528 had strong performance in both categories, scoring 20 in MCQs and 27.5 in TF questions. Google Gemini 2.0 Flash Lite and OpenAI gpt-oss-20b had the highest TF scores, with 35 and 32.5. They also had positive MCQ results. In comparison, Meta Llama 3.1 8B Instruct showed weak MCQ performance with a negative score (−7.5), which significantly lowered its results despite reasonable TF performance (22.5). NVIDIA Nemotron Nano 9B V2 exhibited consistent scores across both question types. Overall, the figure highlights that MCQ performance is the big difference factor, while TF questions show comparatively uniform and higher performance.

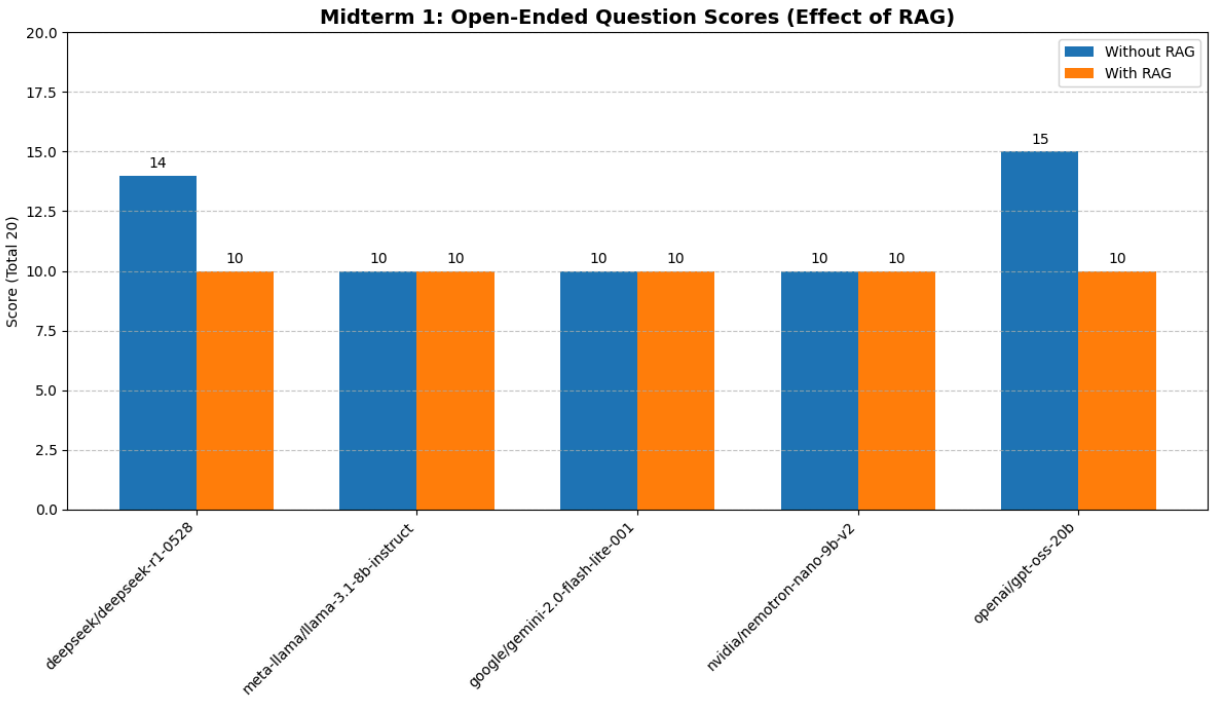

Figure 6: Scores of Open Ended Questions in Midterm 1

Figure 6 shows the scores of open-ended questions in Midterm 1 with and without the use of Retrieval-Augmented Generation. Without it DeepSeek R1-0528 and OpenAI gpt-oss-20b achieved higher scores of 14 and 15. The remaining models scored 10. When RAG was applied, all evaluated models obtained a score of 10 on the open-ended questions. These results show that RAG did not change the open-ended question scores across models.

C. Performance by Topic

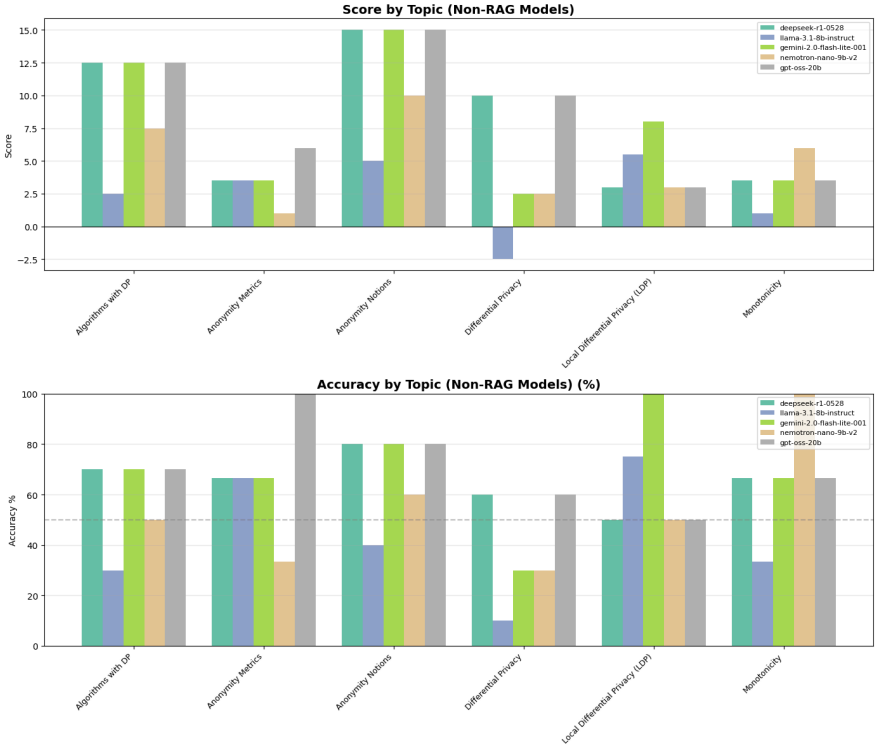

Figure 7 : Performance by Topic MT1 Non-RAG

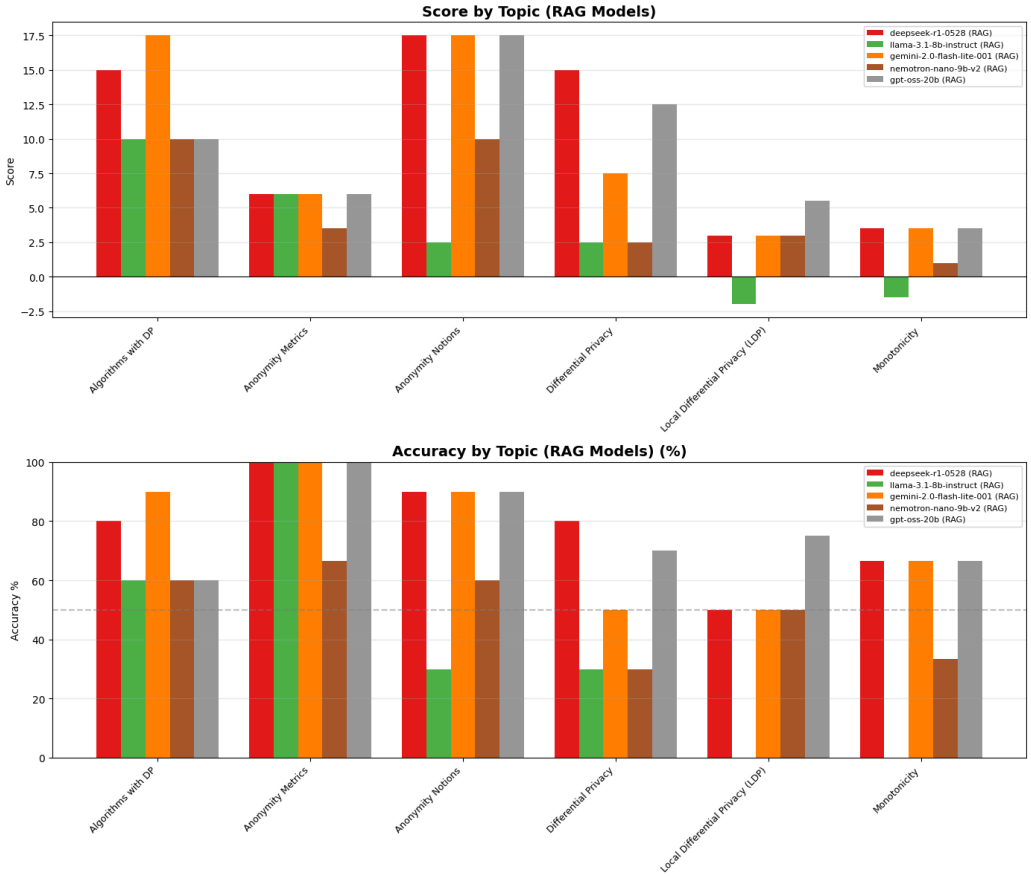

Figure 8: Performance by Topic MT1 RAG

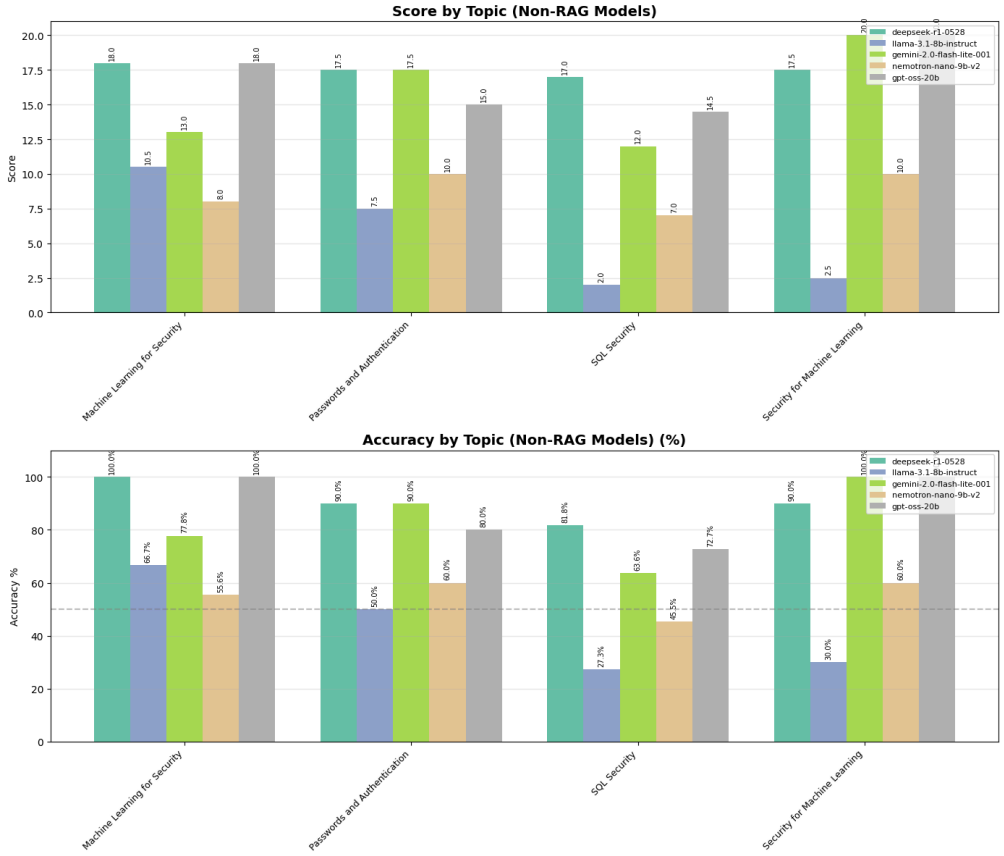

Figure 9 : Performance by Topic MT2 Non-RAG

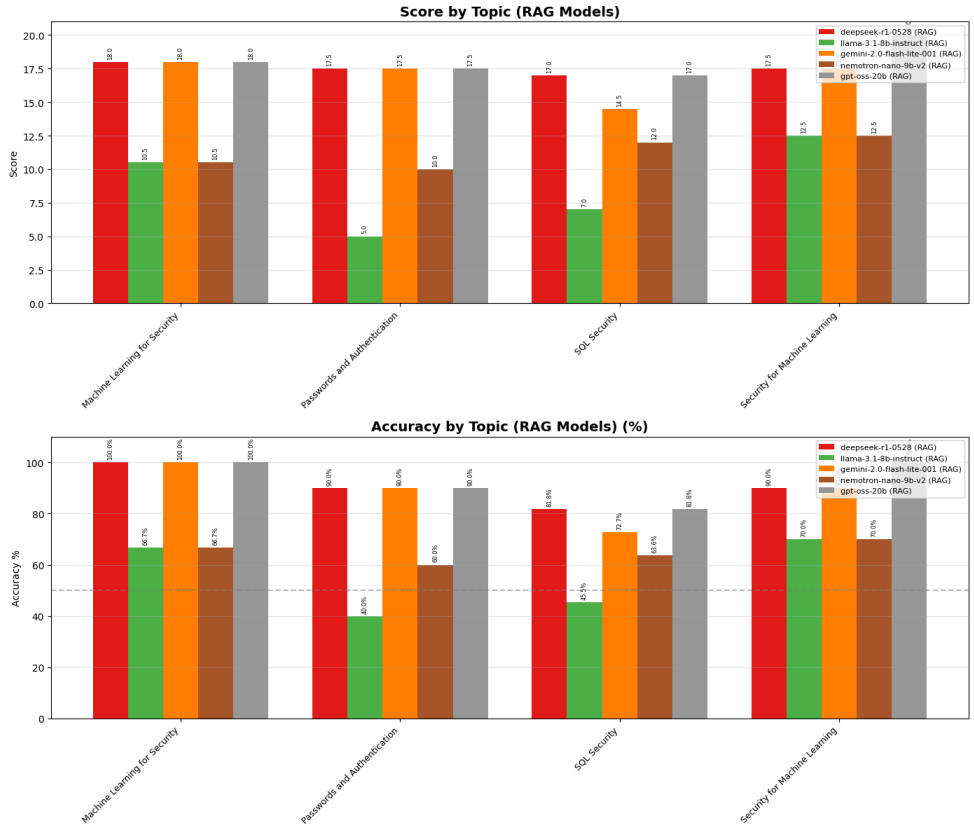

Figure 10 : Performance by Topic MT2 RAG

MT1 Performance by Topic (Non-RAG vs RAG)

As shown in Figure 4 Non-RAG, the models' topic wise performance demonstrates a hierarchy. deepseek-r1-0528 and gpt-oss-20b are the most consistent models across all topics, reaching very high accuracies. DeepSeek achieves 100% on Machine Learning for Security and 90% on Passwords and Authentication, with the weakest topic still strong at 81.8% on SQL Security. Similarly, gpt-oss-20b reaches 100% on Machine Learning for Security and Security for Machine Learning, while weakest topic remains SQL Security (72.7%). Gemini-2.0-flash-lite also performs strongly, 100% in Security for Machine Learning but scoring 63.6% in SQL Security. In contrast, Llama-3.1-8B-Instruct shows weakness across several topics, with its worst performance in SQL Security (27.3%) and Security for Machine Learning (30%), indicating high topic sensitivity. Nemotron-nano-9B-v2 stays moderate across topics, with its weakest again being SQL Security (45.5%), suggesting that SQL-related security reasoning is consistently challenging across smaller models.

When slide-fed prompting is enabled Figure 5, RAG, the topic performance pattern becomes more mixed. For stronger models, RAG generally improves or stabilizes performance in formalized privacy topics: for instance, deepseek-r1-0528 (RAG) 100% in Anonymity Metrics and 90% in Anonymity Notions, and 80% in Algorithms with DP and Differential Privacy. gpt-oss-20b (RAG) also performs strongly in Anonymity Metrics 100% and Anonymity Notions 90%, and improves on LDP to 75%, showing that slide context can support mechanism and definition questions. However, RAG does not uniformly help all models: Llama-3.1-8B-Instruct (RAG) performs well on Anonymity Metrics 100% and improves on Algorithms with DP 60%, but it collapses to 0% on LDP and 0% on Monotonicity, suggesting that irrelevant or confusing retrieved context can harm performance on certain questions. Overall, Figures 4–5 show that RAG can be highly beneficial for top models in DP/anonymity topics, but the effect is model dependent and may introduce failure modes for weaker models, especially on topics requiring careful reasoning.

MT2 Performance by Topic (Non-RAG vs RAG)

In Figure 6 Non-RAG, the Midterm 2 topic again shows that deepseek-r1-0528 performs at a very high level across the board, achieving 100% on Machine Learning for Security, 90% on Passwords and Authentication and Security for Machine Learning, and 81.8% on SQL Security as the weakest topic. gpt-oss-20b follows a similar pattern with 100% in Machine Learning for Security and Security for Machine Learning, while SQL Security remains the lowest topic at 72.7%. Gemini-2.0-flash-lite is strong on authentication and ML-security topics up to 100%, but is again weakest on SQL Security (63.6%). Nemotron-nano-9B-v2 stays mid with its lowest performance in SQL Security (45.5%). Llama-3.1-8B-Instruct is the most topic-sensitive model, with SQL Security at 27.3% and Security for Machine Learning at 30%, showing major weaknesses on database-oriented and security-ML questions.

With RAG enabled Figure 7, the strongest models largely remain stable rather than dramatically improving. DeepSeek stays identical across topics (e.g., still 100% on Machine Learning for Security and 81.8% on SQL Security), where the model's internal knowledge is sufficient. gpt-oss-20b, however, shows a clearer benefit from slide context, improving Passwords and Authentication from 80% to 90% and SQL Security from 72.7% to 81.8%, and maintaining 100% in security-ML topics. The most noticeable improvements occur for weaker models such as: Llama-3.1-8B-Instruct increases SQL Security from 27.3% to 45.5% and Security for Machine Learning from 30% to 70%, indicating that retrieved course material can effectively help. Nemotron-nano-9B-v2 also improves substantially in SQL Security (45.5% to 63.6%) and Security for Machine Learning (60% to 70%). Overall, Figures 6–7 indicate that RAG is most valuable for smaller, weaker models in Midterm 2, especially in SQL security and security-for-ML topics, while top-tier models show smaller improvement because they already have a high baseline.

D. Performance by Difficulty

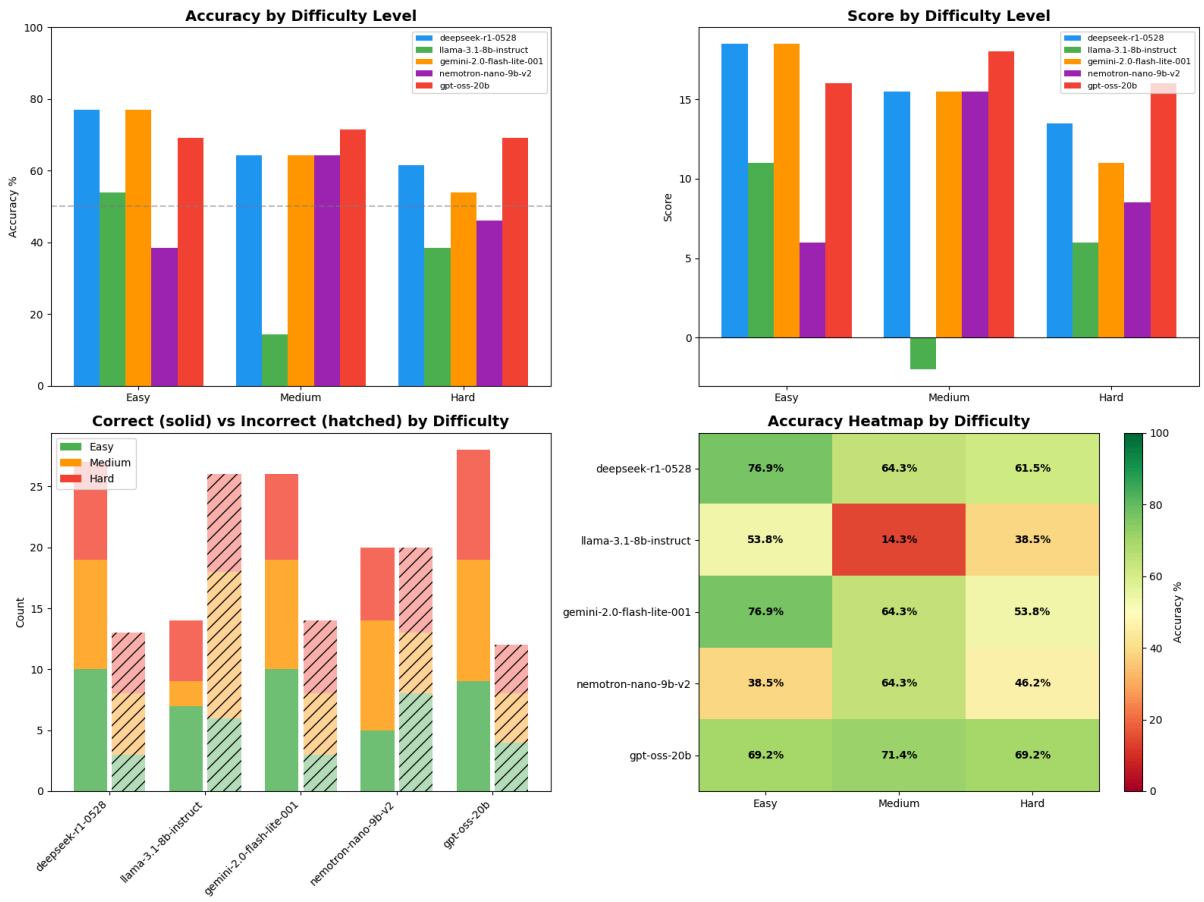

Figure 11: Midterm 1 No Rag Results with Different Graphs

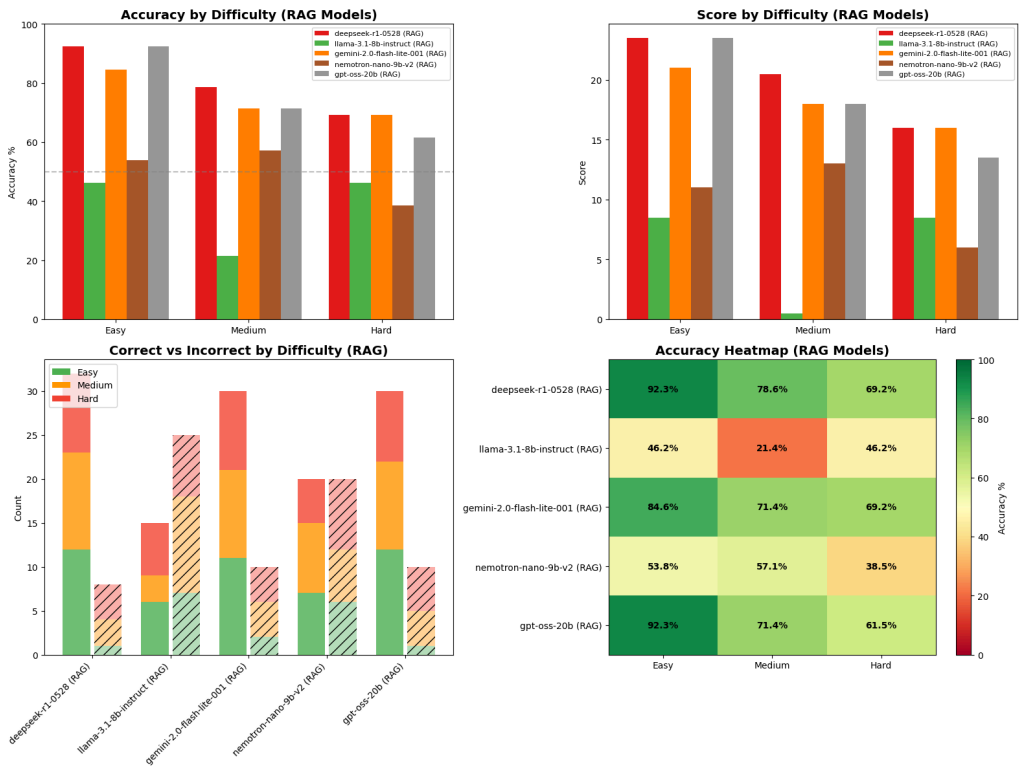

Figure 12: Midterm 1 RAG additional graphs

Figure 11 and Figure 12 present the performance of the evaluated LLMs across three difficulty levels (easy, medium, hard) under non-RAG and RAG conditions, respectively. Across almost all models and both settings, medium-difficulty questions emerge as the most consistently well-performing category, while hard questions remain the primary source of performance degradation especially for smaller models.

In the non-RAG setting Figure 11, a clear pattern is that stronger models maintain high accuracy across all difficulty levels. deepseek-r1-0528 performs best on medium questions 94.4%, while still maintaining high accuracy on hard 90.0% and easy 83.3% questions, indicating stable reasoning capability across difficulty tiers. Similarly, gpt-oss-20b achieves its strongest performance on easy questions 91.7%, while remaining highly consistent on medium 88.9% and hard 80.0%, suggesting robust generalization even as difficulty increases. gemini-2.0-flash-lite-001 follows a similar trend, peaking on medium questions 88.9% and showing only a moderate drop on hard questions 80.0%.

In contrast, the smaller models exhibit sharper sensitivity to difficulty. llama-3.1-8b-instruct struggles across all levels, with particularly weak performance on easy questions 33.3% and hard questions 40.0%, indicating that even surface-level questions are not reliably solved. nemotron-nano-9b-v2 shows an unusual pattern: while it performs best on hard questions 70.0%, it performs significantly worse on easy questions 33.3%, suggesting instability and inconsistent calibration across difficulty levels.

Under the RAG setting (Figure 12), several notable shifts occur. For the stronger models, RAG tends to improve or stabilize medium-difficulty performance while having mixed effects on hard questions. deepseek-r1-0528 improves its medium accuracy from 94.4% to 100.0%, while its hard accuracy decreases slightly from 90.0% to 80.0%, indicating that RAG helps most in mid-complexity reasoning but may introduce minor distraction effects on harder problems. gpt-oss-20b shows the most balanced RAG gains, improving medium accuracy from 88.9% to 94.4% and hard accuracy from 80.0% to 90.0%, demonstrating that slide-fed context can meaningfully support higher-difficulty reasoning for large models. gemini-2.0-flash-lite-001 remains highly consistent, with improvements in medium (88.9% to 94.4%) and stable performance on hard (80.0%), reinforcing its reliability under contextual augmentation.

For the smaller models, RAG produces more volatile and sometimes negative effects. llama-3.1-8b-instruct improves substantially on easy questions (33.3% to 66.7%) and medium questions (50.0% to 66.7%), but collapses on hard questions (40.0% to 20.0%), suggesting that retrieval noise disproportionately harms performance on already difficult reasoning tasks. nemotron-nano-9b-v2 shows moderate RAG gains on easy (33.3% to 66.7%) and medium (61.1% to 72.2%) questions, but a decline on hard (70.0% to 50.0%), again indicating that RAG can help compensate for missing background knowledge but may interfere with high-complexity reasoning when retrieval quality is imperfect.

Overall, Figures 11–12 indicate three core trends. First, medium-difficulty questions are the most reliable performance band across models and settings, suggesting that they best align with the typical reasoning depth of modern LLMs. Second, hard questions remain the primary failure point, especially for smaller models and under RAG, where retrieval noise can amplify errors. Third, RAG provides the most consistent benefit for medium-difficulty questions and for larger models (notably gpt-oss-20b), while its effect on hard questions is highly model-dependent. These findings reinforce the conclusion that difficulty level is a major moderator of LLM reliability and that contextual augmentation should be applied selectively rather than uniformly across all question complexities.

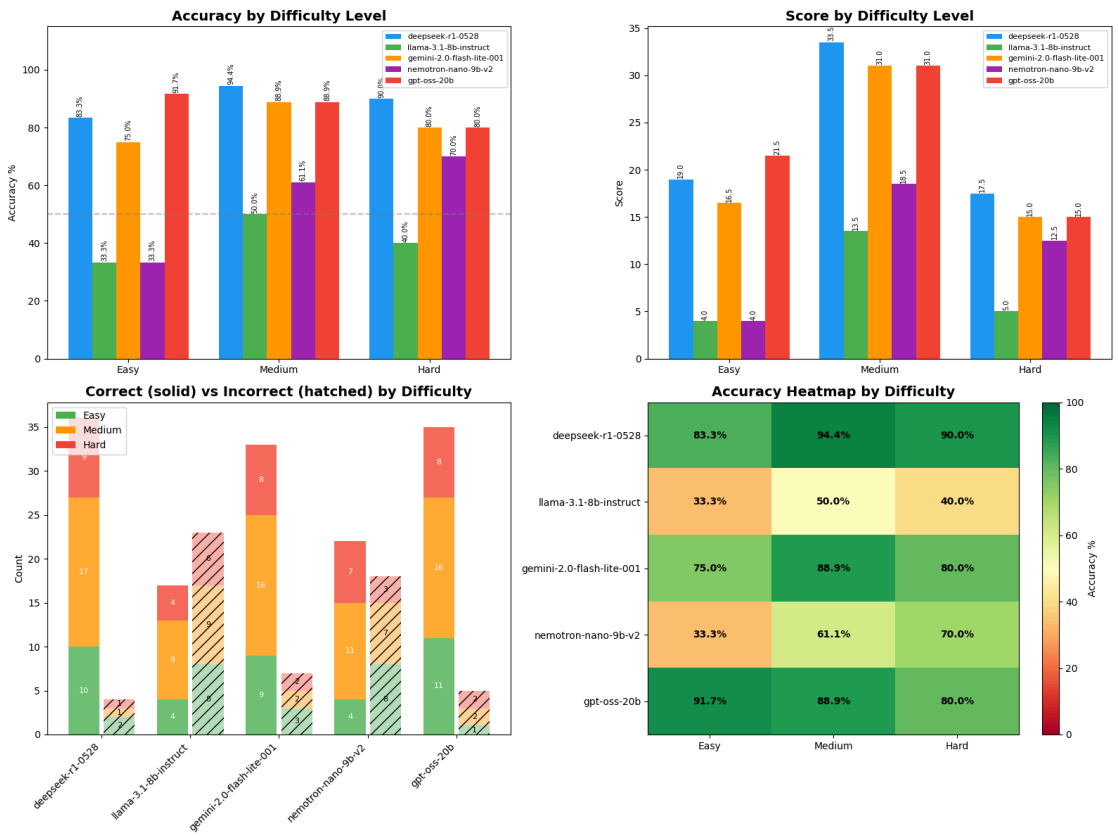

Figure 13: Midterm 2 Non-Rag by Difficulty

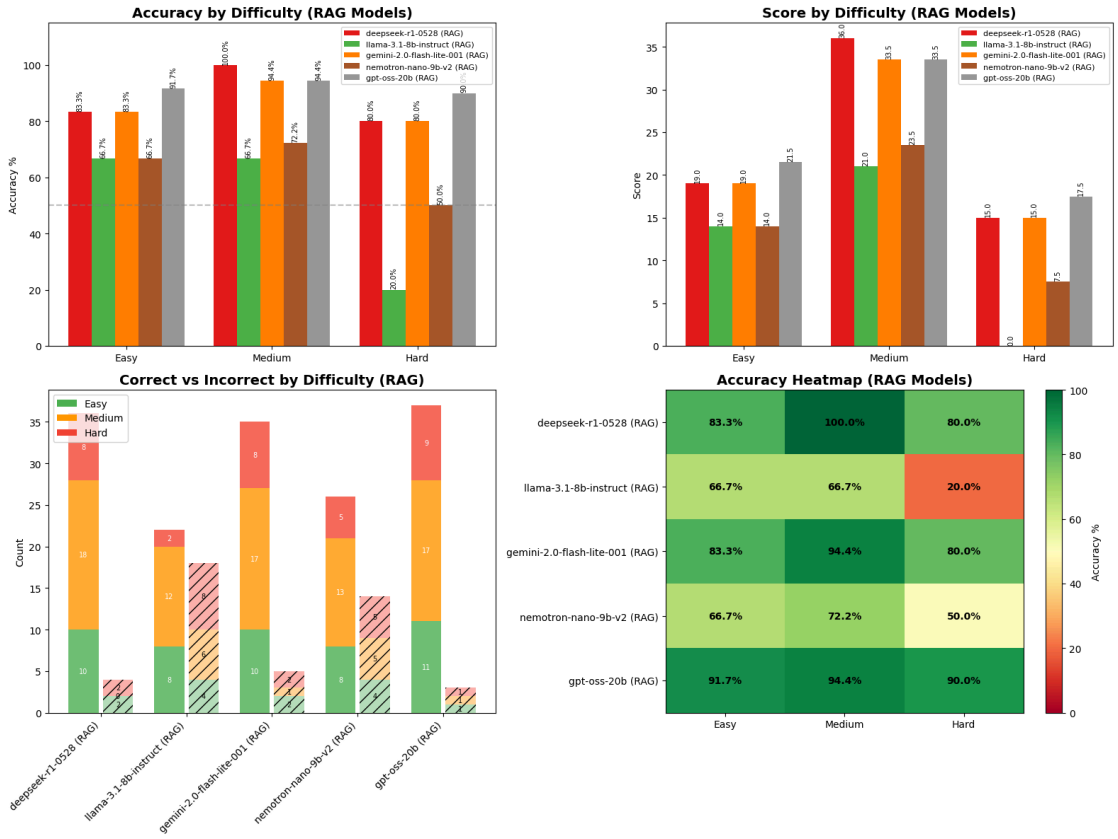

Figure 14: Midterm 2 Rag with difficulty indicators

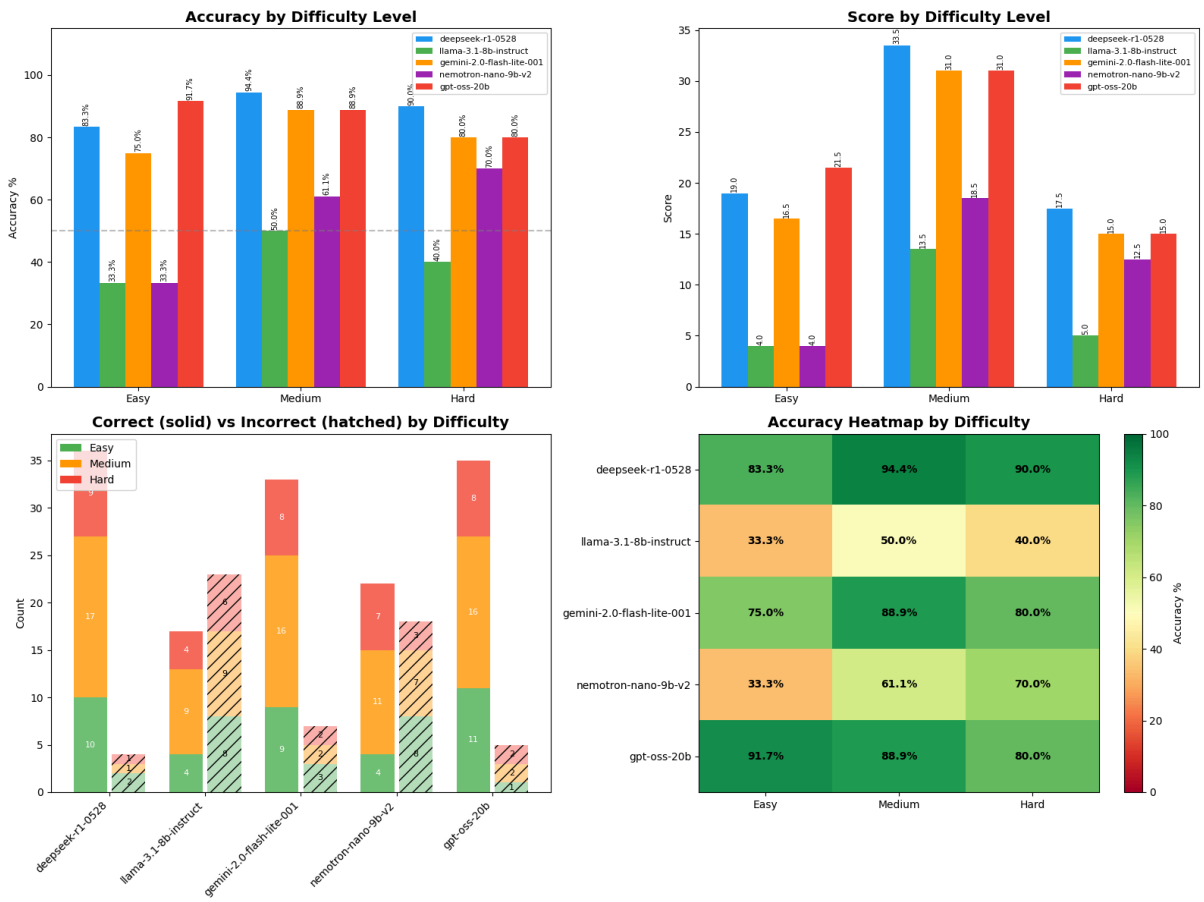

Figure 15: Midterm 1, RAG vs No RAG

E. Effect of RAG

The integration of Retrieval-Augmented Generation (RAG) demonstrated a varied impact across the evaluated models and exam formats. While RAG generally served as a powerful enhancement tool, particularly for domain-specific true-false or multiple-choice questions, it was not universally beneficial. The results indicate a "trade-off" where retrieval can boost factual accuracy but may occasionally introduce noise that degrades performance in open-ended reasoning tasks.

Mixed Reliability: RAG proved to be highly effective for standardized, fact-based questions Midterm 1, where it significantly boosted scores for models like Deepseek. In open-ended scenarios, specifically Midterm 1 and Midterm 2, RAG caused performance regressions for top-tier models like OpenAI GPT-OSS-20b and Llama-3.1-8b, suggesting that irrelevant context can confuse the model and disrupt its internal reasoning chain.

Context Sensitivity: Smaller models (Llama, Nemotron) benefited most consistently from RAG in the multiple-choice sections of Midterm 2 (gaining +10-12.5%), as they relied on the external knowledge to bridge gaps. Conversely, larger models with strong internal knowledge (Deepseek, GPT-OSS) showed diminishing returns or even negative impacts when the retrieved context did not align perfectly with the nuanced requirements of open-ended questions.

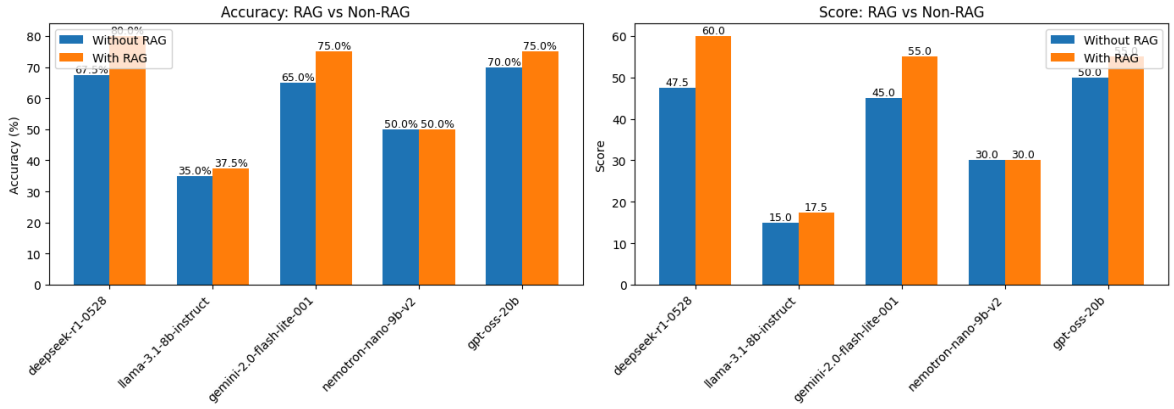

Figure 16: Midterm 1, RAG vs No RAG

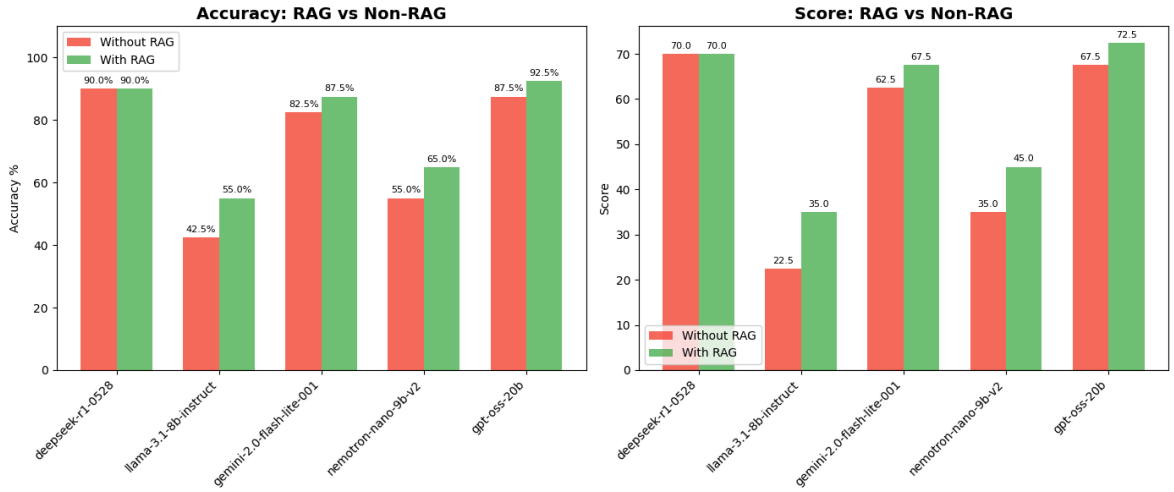

Figure 17: Midterm 2, RAG vs No RAG

Overall, in midterm 2 GPT-OSS-20b emerged as the highest-performing model overall, leveraging RAG to boost its accuracy from 87.5% to 92.5%. Deepseek-r1-0528 demonstrated exceptional baseline capability, achieving 90.0% accuracy in both configurations. This suggests its internal knowledge base was already sufficient for the tested dataset, reaching a saturation point where RAG was not strictly necessary but still stable.

The most dramatic improvements were observed in models with lower initial baselines, highlighting RAG's ability to effectively bridge knowledge gaps. Llama-3.1-8b-instruct recorded the largest improvement.

RAG effectiveness comparison between Midterm 1 and Midterm 2:

Comparing the RAG improvement metrics between the two midterms reveals distinctly different behaviors across the models. RAG proved more critical for Midterm 1, delivering larger average performance boosts for the top-tier models (Deepseek and Gemini gained +12.5% and +10.0% respectively). In contrast, Midterm 2 showed a "saturation effect" for these stronger models, where baselines were already so high (e.g., Deepseek at 90%) that RAG offered little to no extra value. Conversely, the smaller models (Llama and Nemotron) benefited significantly more from RAG in Midterm 2 (gaining +10-12.5%) compared to their poor performance in Midterm 1. This suggests that Midterm 1 likely required more specific, retrievable facts that the larger models could leverage effectively, whereas Midterm 2 might have relied more on generalizable knowledge that allowed even smaller models to outperform their baselines given the right context.

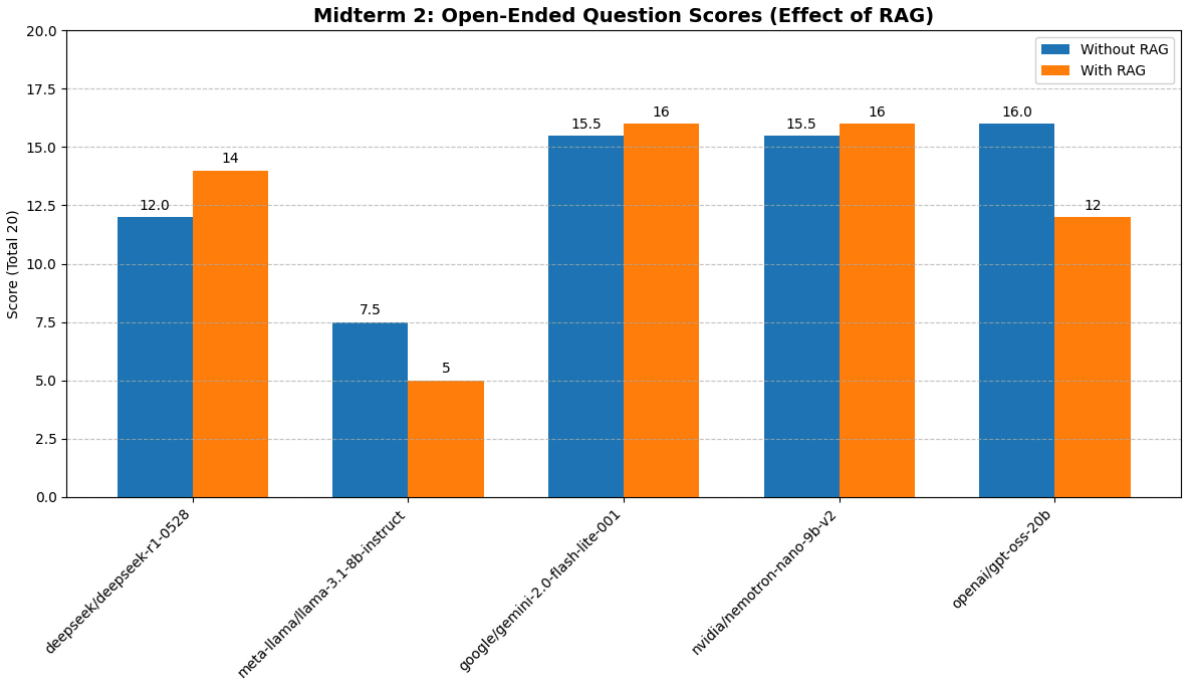

Figure 18: Midterm 2 Open-Ended, No RAG vs RAG

In summary, while RAG is generally a powerful tool, this specific test case highlights that it is not a universal fix-all; for open-ended, reasoning-heavy tasks, it can sometimes degrade performance by interfering with a model's internal logic.

VII. Discussion

The results indicate that while modern LLMs demonstrate the potential to pass COMP 430-style exams, their performance remains highly inconsistent. The models showed clear strengths in recognizing factual patterns and accurately answering direct conceptual questions. However, robust technical capability is constrained by significant weaknesses:

- Models frequently failed to maintain coherent logic across sequential problem-solving stages.
- Precise execution of specific algorithms (e.g., anonymization steps) often lacked accuracy.
- Models struggled to differentiate between closely related security or privacy concepts in nuanced scenarios.

The mixed impact of RAG suggests that access to course materials is helpful but not sufficient to guarantee correctness. Careful retrieval filtering and prompt design are necessary to avoid misleading context injection. Even though there were no bad effects of RAG for True/False and MCQ questions, overall there was a downgrade of the performance for the open-ended questions.

From an educational perspective, these findings imply that LLMs may be useful as supplementary study tools but should not be trusted as authoritative sources for high-stakes learning in privacy and security.

VIII. Limitations

This study was constrained by budget limitations, restricting the evaluation to a subset of open-weights and cost-effective models while excluding the most advanced, high-cost proprietary LLMs. Additionally, the dataset represents a small sample of the full course curriculum, and the reliance on manual rubric-based grading for open-ended questions introduces an element of inherent subjectivity to the evaluation.

IX. Conclusion

This project presented a systematic benchmarking study of modern LLMs on COMP 430–level privacy and security exam questions. While some models achieve passing-level scores, none demonstrate consistent mastery across all topics and difficulty levels. The results highlight both the potential and the limitations of using LLMs as educational tools in high-stakes technical domains. Incorporating course materials via RAG can improve performance in certain contexts, but it also introduces new sources of error. Overall, LLMs should be viewed as supportive learning aids rather than reliable substitutes for deep conceptual understanding in data privacy and security. We successfully completed the project and showed our results as we wrote in our project proposal.

References

[1] K. Bittle and O. El-Gayar, “Generative AI and Academic Integrity in Higher Education: A Systematic Review and Research Agenda,” Information, vol. 16, no. 4, p. 296, 2025.

[2] N. Rane et al., “Artificial Intelligence, ChatGPT, and the New Cheating Dilemma: Strategies for Academic Integrity,” 2024.

[3]“DeepSeek-R1-0528 Throughput API | Together AI,” *Together.ai*, 2020. https://www.together.ai/models/deepseek-r1-0528-throughput

[4]“meta-llama/Llama-3.1-8B-Instruct · Hugging Face,” *Huggingface.co*, Sep. 25, 2024. https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct

[5]“GPT-OSS vs Gemini 2.0 Flash-Lite - Detailed Performance & Feature Comparison,” *DocsBot AI*, 2024. https://docsbot.ai/models/compare/gpt-oss/gemini-2-0-flash-lite

[6]“Nemotron Nano 9B V2 vs gpt-oss-20b - AI Model Comparison | OpenRouter,” *OpenRouter*, 2026. https://openrouter.ai/compare/nvidia/nemotron-nano-9b-v2/openai/gpt-oss-20b

[7]R. Bathgate, “Everything you need to know about OpenAI’s new open weight AI models, including price, performance, and where you can access them,” *IT Pro*, Aug. 06, 2025. https://www.itpro.com/technology/artificial-intelligence/openai-open-weight-ai-models